Lecture 22
Significance and P-values

Review: which interval?

a proportion p

$\hat{p} \pm z \sqrt{\hat{p}(1 - \hat{p}) / n}$, when $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$

a mean μ, σ known

$\bar{x} \pm z \sigma / \sqrt{n}$

a mean μ, σ unknown

$\bar{x} \pm t s / \sqrt{n}$, with $df = n - 1$

when the t interval works

the population is roughly normal, or $n \geq 30$

Data: seal whiskers

Ginter et al. (2012) measured the longest undamaged whisker of 92 seals and sea lions from 11 species, to study how whisker shape evolved.

X = the total length of a seal's longest whisker, in mm

SPECIES	SEALS (n)	MEAN LENGTH (mm)	SD (mm)
Harp seal	16	80.6	10.8
Harbor seal	15	86.5	10.5
Ringed seal	16	79.6	10.9
Spotted seal	9	91.2	7.9
Gray seal	8	73.8	12.5
…	…	…	…

One whisker per seal; only each species' n, mean and SD were published

Example: spotted seal whiskers

Estimate μ, the mean whisker length of all spotted seals, from the 9 that were measured.

n = 9; $\bar{x}$ = 91.2 mm; s = 7.9 mm

(a) z or t? How many degrees of freedom?
(b) What is the standard error?
(c) What is the 95% confidence interval for μ?

(a) t, because σ is unknown; df = 9 – 1 = 8, t = 2.306
(b) 7.9 / √9 ≈ 2.63 mm
(c) 91.2 ± 2.306(2.63) = 91.2 ± 6.06: 85.1 to 97.3 mm

Choosing a sample size

margin of error m

the ± part of an interval: $m = z \times SE$, so a larger n gives a smaller m

n for a proportion

$n = z^2\, p^*(1 - p^*) / m^2$, with $p^* = 0.5$ if there is no earlier guess

n for a mean

$n = (z\, \sigma / m)^2$, with σ from an earlier study

always round up

n = 59.1 means 60: rounding down misses the target m

Example: planning the next study

Before collecting data, decide how many to measure so the 95% interval is as narrow as you need.

carriers: m = 0.03, p* = 0.5; whiskers: m = 2 mm, σ ≈ 7.9 mm

(a) How many people for the proportion of *2 carriers?
(b) How many spotted seals for the mean whisker length?
(c) What happens to n if we want half the margin of error?

(a) $1.96^2\,(0.5)(0.5) / 0.03^2 = 1067.1$ → 1,068 people
(b) $(1.96 \times 7.9 / 2)^2 = 59.9$ → 60 seals
(c) 4 times as large: m is squared on the bottom

Practice: how many students?

Campus dining wants to estimate p, the proportion of students who skip breakfast, to within 4 percentage points with 90% confidence.

m = 0.04; 90% confidence; last year's survey: 30% skip breakfast

(a) How many students should they survey, using last year's 30%?
(b) How many if there were no earlier survey?
(c) Why is (b) larger?

(a) $1.645^2\,(0.3)(0.7) / 0.04^2 = 355.2$ → 356 students
(b) $1.645^2\,(0.5)(0.5) / 0.04^2 = 422.8$ → 423 students
(c) $p^*(1 - p^*)$ is largest at $p^* = 0.5$, so 0.5 is the safe guess

Can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. In a blind test, you pour one of the two at random, 8 times, and they name each cup.

8 of 8 cups named correctly

(1) Can they really tell, or were they just lucky? How could we decide?
(2) If they were only guessing, how likely is 8 for 8?
(3) What are the two ways to explain such a small probability?

(2) $(1/2)^8 = 1/256 \approx 0.004$
(3) Either something rare happened by chance, or "just guessing" is wrong
How rare is rare enough is our choice; by convention, below 0.05

Significance tests

statistically significant

decidedly not due to ordinary variation: not chance, not a coincidence

significance test

uses data to decide whether a result is statistically significant

null hypothesis H_0

"no effect": nothing interesting is happening, e.g. they are just guessing

alternative hypothesis H_A

"an effect": what we suspect, e.g. they can taste the difference

Writing the hypotheses

θ and θ_0 ("theta" and "theta-naught")

θ is the parameter (p or μ); θ_0 is the value H_0 claims, e.g. $p_0 = 0.5$

$H_0: \theta = \theta_0$

H_0 always has the = sign, and is about a parameter, never $\hat{p}$ or $\bar{x}$

$H_A: \theta > \theta_0$, $\theta < \theta_0$ or $\theta \neq \theta_0$

upper-tailed, lower-tailed (one-sided) or two-sided; set before seeing data

Example: stating hypotheses

For each question, write H_A first (the suspicion), then H_0 (no effect). Give θ_0 and say which tail.

p = a population proportion; μ = a population mean

(a) Your roommate names Coke or Pepsi. Do they do better than guessing?
(b) A coin is spun on a table. Does it favor heads or tails?
(c) A chip bag says 28 g. Do the bags hold less than that on average?

(a) H_A: $p > 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, upper-tailed (p = chance of a right cup)
(b) H_A: $p \neq 0.5$ H_0: $p = 0.5$ $p_0 = 0.5$, two-sided (p = chance of heads)
(c) H_A: $\mu < 28$ g H_0: $\mu = 28$ g $\mu_0 = 28$ g, lower-tailed

Test statistics and P-values

test statistic z_{obs}

z_{obs} = (estimate − θ_0) / SE: how many SEs the estimate is from θ_0

P-value

a result as extreme as ours, or more, if H_0 is true; upper-tailed: $P(Z \geq z_{obs} \mid H_0 \text{ true})$

significance level α

the cutoff, chosen before collecting data; usually $\alpha = 0.05$

the decision

P-value < α: reject H_0 (significant); P-value ≥ α: do not reject H_0

Example: taste-test P-values

A second friend makes the same claim and takes a 6-cup test. Test H_0: $p = 0.5$ against H_A: $p > 0.5$ at $\alpha = 0.05$.

roommate: 8 of 8 correct; friend: 4 of 6 correct

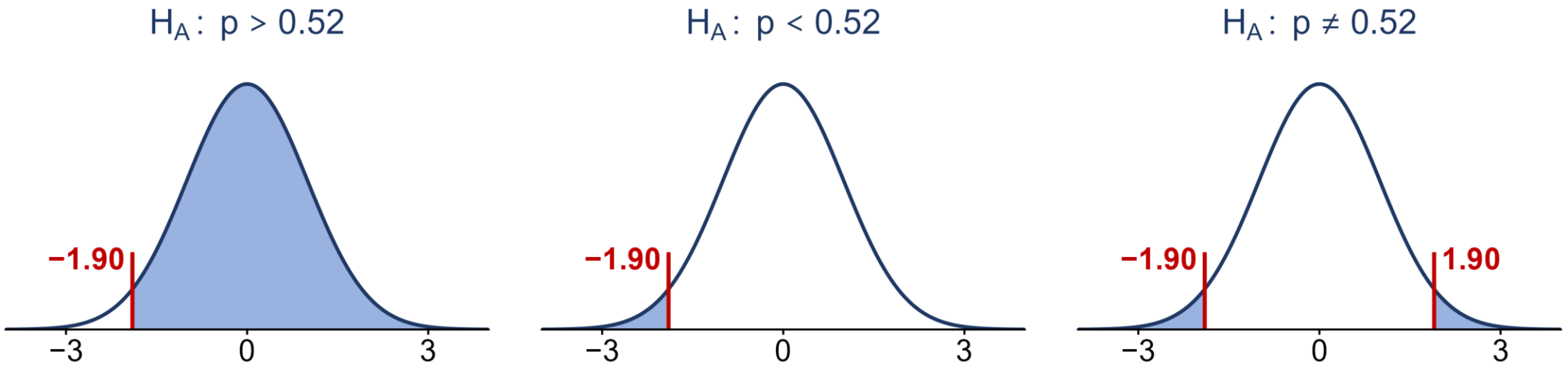
(a) If H_0 is true, what is the P-value for each?
(b) What is the decision for each?
(c) If 256 people took the 8-cup test by pure guessing, how many would go 8 for 8?

(a) roommate: $P(8 \text{ of } 8 \mid H_0 \text{ true}) = 0.5^8 \approx 0.004$
friend: $P(4 \text{ or more of } 6 \mid H_0 \text{ true}) = 22/64 \approx 0.34$
(b) roommate: $0.004 < 0.05$, reject H_0; friend: $0.34 \geq 0.05$, do not reject H_0
(c) About 1: a small P-value means rare luck or a false H_0, and we can't be sure which

Which tail is the P-value?

Polls put Candidate X at 52%. Our poll of 1,000 likely voters finds 49.0%, so testing H_0: p = 0.52 gives z_{obs} = −1.90.

What is the P-value under each H_A? Which ones reject H_0 at α = 0.05?



upper: P(Z ≥ −1.90) = 0.971 lower: P(Z ≤ −1.90) = 0.029 two: 2 × 0.029 = 0.057

Only the lower-tailed test rejects H_0: H_A must be set before seeing the data

Data: 350,757 coin flips

A physics model predicts a flipped coin lands on the side it started on about 51% of the time. Bartoš et al. (2025) had 48 people flip coins to test it.

X = 1 if a flip lands on the side it started on, and 0 otherwise

who flipped	**flips**
students, for a thesis project	75,036
coin-flipping marathons (up to 12 hours)	203,440
volunteers recruited online	72,281
total	**350,757**

Each person recorded the side facing up before and after every flip, and filmed it

Example: do coins land the way they started?

Test whether flips land on their starting side more than half the time, at $\alpha = 0.05$. The model predicted the direction before any flips were made.

178,079 of n = 350,757 flips landed on the side they started on

(a) What are p_0, H_0 and H_A? Which tail?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision?
(d) 50.8% versus 50%: is that a big difference?

(a) $p_0 = 0.5$; $H_0: p = 0.5$ $H_A: p > 0.5$; upper-tailed
(b) $\hat{p} = 178{,}079 / 350{,}757 \approx 0.5077$; $SE = \sqrt{(0.5 \times 0.5 / 350{,}757)} \approx 0.000844$
$z_{obs} = (0.5077 - 0.5) / 0.000844 \approx 9.1$
(c) $P(Z \geq 9.1 \mid H_0 \text{ true}) \approx 0$, off the z-table: less than 0.05, so reject H_0
(d) No: about 1 extra in 130 flips. Significant means "not chance", not "big"

The five steps of a significance test

1. assumptions, 2. hypotheses

a random sample (plus size or shape checks); H_0: $\theta = \theta_0$ and H_A

3. test statistic

z_{obs} = (estimate – θ_0) / SE, computed as if H_0 were true

4. P-value

the area beyond z_{obs} in the tail or tails H_A points to, if H_0 were true

5. decision

P-value < α: reject H_0; otherwise do not reject; significant ≠ important